

**Table 1: Descriptive Statistics for 7 Targets & Engineered Descriptors (N = 115)**

| Feature / Property     | Unit          | Count   | Mean   | Std Dev | Min    | Median | Max      |
|------------------------|---------------|---------|--------|---------|--------|--------|----------|
| E_ads_DFT_eV           | eV            | 115.000 | -2.938 | 1.065   | -7.102 | -2.946 | -0.819   |
| band_gap               | eV            | 115.000 | 1.188  | 1.374   | 0.000  | 0.858  | 7.356    |
| formation_energy       | eV/atom       | 115.000 | -1.002 | 0.938   | -3.642 | -0.845 | 0.821    |
| energy_hull            | eV/atom       | 115.000 | 1.115  | 1.044   | 0.000  | 0.954  | 5.107    |
| bulk_modulus           | GPa           | 115.000 | 57.789 | 60.852  | 5.000  | 37.680 | 419.280  |
| shear_modulus          | GPa           | 115.000 | 32.365 | 52.523  | 1.320  | 19.520 | 475.470  |
| young_modulus          | GPa           | 115.000 | 80.056 | 119.018 | 3.873  | 52.595 | 1035.127 |
| work_function_eV       | eV            | 115.000 | 4.949  | 0.947   | 3.190  | 4.950  | 9.302    |
| mean_atomic_mass       | amu           | 115.000 | 69.022 | 39.640  | 12.011 | 63.403 | 204.348  |
| mean_electronegativity | Pauling       | 115.000 | 2.502  | 0.487   | 1.295  | 2.456  | 3.682    |
| pugh_ratio             | dimensionless | 115.000 | 2.477  | 2.237   | 0.882  | 1.846  | 14.826   |
| poisson_ratio          | dimensionless | 115.000 | 0.280  | 0.067   | 0.089  | 0.271  | 0.467    |
| hardness_gpa           | GPa           | 115.000 | 4.887  | 7.931   | 0.199  | 2.948  | 71.796   |